

Report of Major Project Phase-II (SPW818EC)

On

**"SCALABLE APPROACH FOR POWER DROOP
REDUCTION DURING SCAN BASED LOGIC BIST"**

Submitted in partial fulfillment of the requirements for the award of the

degree of

BACHELOR OF ENGINEERING

In

ELECTRONICS AND COMMUNICATION ENGINEERING

By

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Under the esteemed guidance of

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Department of Electronics and Communication Engineering
STANLEY COLLEGE OF ENGINEERING AND TECHNOLOGY FOR
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Chapel road, Hyderabad – 500001

(Affiliated to Osmania University & Approved by AICTE) (All eligible UG
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2025-2026



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CERTIFICATE

This is to certify that the project titled “**Scalable approach for power droop reduction during Scan based Logic BIST**” is a bonafide work carried over by **Ms.G.Hima Bindhu (160622735088)** in partial fulfillment of the requirements for the award of the Degree Bachelor of Engineering in Electronics and Communication Engineering from Stanley College of Engineering and Technology for Women (Autonomous), Hyderabad affiliated to Osmania University, during the VIII semester of their B.E course during the Academic Year 2025-2026.

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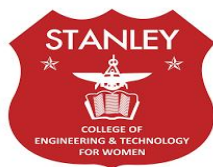
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PSO2: Adaptability to Productive Environment: To be well equipped with Management skills, interdisciplinary and modern technologies.



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SUSTAINABLE DEVELOPMENT GOALS

SD G No.	Goal Name	Explanation to SDG
1	No Poverty	End poverty in all its forms everywhere by ensuring access to basic needs and resources.
2	Zero Hunger	End hunger, achieve food security, improve nutrition, and promote sustainable agriculture.
3	Good Health and Well-being	Ensure healthy lives and promote well-being for all people at all ages.
4	Quality Education	Ensure inclusive and equitable quality education and promote lifelong learning opportunities.
5	Gender Equality	Achieve gender equality and empower all women and girls.
6	Clean Water and Sanitation	Ensure availability and sustainable management of water and sanitation for all.
7	Affordable and Clean Energy	Ensure access to affordable, reliable, sustainable, and modern energy.

8	Decent Work and Economic Growth	Promote sustained, inclusive economic growth, productive employment, and decent work.
9	Industry, Innovation and Infrastructure	Build resilient infrastructure, promote inclusive industrialization, and foster innovation.
10	Reduced Inequalities	Reduce inequality within and among countries.
11	Sustainable Cities and Communities	Make cities and human settlements inclusive, safe, resilient, and sustainable.
12	Responsible Consumption and Production	Ensure sustainable consumption and production patterns.
13	Climate Action	Take urgent action to combat climate change and its impacts.
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16	Peace, Justice and Strong Institutions	Promote peaceful and inclusive societies, provide access to justice, and build effective institutions.
17	Partnerships for the Goals	Strengthen global partnerships to achieve sustainable development goals.



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- WK1.** A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
- WK2.** Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
- WK3.** A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
- WK4.** Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
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ACKNOWLEDGEMENT

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G.Hima Bindhu (16062273588)

ABSTRACT

Logic Built-In Self-Test (LBIST) is widely used for at-speed testing of digital circuits; however, it often introduces significant power consumption during scan operations, leading to power droop issues. These power fluctuations can cause timing violations, false failures, and reduced circuit reliability.

This project presents a power droop reduction technique during scan-based LBIST by incorporating segmented scan chain architecture and controlled scan activation. The proposed system utilizes an LFSR (Linear Feedback Shift Register) for pseudo-random test pattern generation and a MISR (Multiple Input Signature Register) for response compaction. A clock divider is employed to generate a slower scan clock, reducing switching activity.

To further mitigate power droop, the scan chain is divided into segments, and a segment counter and decoder selectively activate scan segments. The droop controller logic ensures that only a portion of the scan chain is active at a time, thereby reducing simultaneous switching and peak power demand.

The design is implemented in Verilog HDL and simulated using Xilinx Vivado. The results demonstrate reduced switching activity and improved power stability during testing without compromising fault coverage. This approach provides an efficient and scalable solution for low-power VLSI testing.

Keywords

Logic Built-In Self-Test (LBIST), Power Droop Reduction, Segmented Scan Chain, Low Power Testing, LFSR, MISR, VLSI Testing.

Expected PO's

PO1 – Engineering Knowledge

PO2 – Problem Analysis

PO3 – Design / Development of Solutions

PO4 – Investigation of Complex Problems

PO5 – Modern Tool Usage

PO7 – Environment and Sustainability

PO9 – Individual and Team Work

Expected PSO's

PSO1 – Appertian to Communication and Automation Principles

PSO2 – Adaptability to Productive Environment

Expected SDG's

SDG 7 – Affordable and Clean Energy

SDG 9 – Industry, Innovation, and Infrastructure

SDG 12 – Responsible Consumption and Production

SDG 13 – Climate Action

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
LBIST	Logic Built-In Self-Test
BIST	Built-In Self-Test
LFSR	Linear Feedback Shift Register
MISR	Multiple Input Signature Register
DFT	Design For Testability
PRPG	Pseudo-Random Pattern Generator
RTL	Register Transfer Level
FPGA	Field Programmable Gate Array
VLSI	Very Large Scale Integration
HDL	Hardware Description Language

CHAPTER 1

INTRODUCTION

1.1 Aim of the project

A power-efficient RTL architecture is designed, implemented, and simulated in Verilog to mitigate power droop during scan-based Logic Built-In Self-Test (LBIST) in digital VLSI circuits. Power droop, caused by excessive simultaneous switching activity in scan chains during test operations, remains a major challenge in modern integrated circuits, often leading to timing violations, reduced reliability, and false test failures. This issue is addressed through architectural enhancements that effectively control switching activity while maintaining low hardware overhead.

A segmented scan chain architecture is employed, where only a portion of the scan chain is activated at a time instead of enabling all flip-flops simultaneously. This selective activation is managed using a segment counter and a one-hot decoder, ensuring that switching transitions are evenly distributed across multiple clock cycles, thereby reducing peak current demand. Additionally, a droop controller dynamically regulates the scan enable signal to suppress unnecessary toggling and further stabilize power consumption.

To enhance power efficiency, a clock divider is incorporated to reduce the operating frequency during scan operations, which lowers instantaneous current demand and improves overall power integrity. An 8-bit Linear Feedback Shift Register (LFSR) is used for efficient pseudo-random test pattern generation, while an 8-bit Multiple Input Signature Register (MISR) compacts the output responses into a signature for verification, ensuring effective fault detection with minimal overhead.

The complete architecture is validated through functional simulation using Xilinx Vivado, confirming correct module operation and demonstrating its effectiveness in reducing power droop. The design provides a scalable, low-overhead, and reliable LBIST framework that enhances power stability without significantly compromising test efficiency, making it suitable for modern low-power, high-performance VLSI systems.

1.2 Objectives of project

The objectives focus on developing a power-efficient scan-based LBIST architecture by incorporating design techniques that effectively reduce switching activity during test operations. A key objective is to design a segmented scan chain architecture in which only one segment is activated at a time, instead of enabling all scan flip-flops simultaneously. This approach helps in minimizing excessive switching transitions, thereby reducing peak power consumption and mitigating power droop issues. To enable this controlled activation, a one-hot segment decoder is designed along with droop control logic, which dynamically manages the scan enable signals across different scan segments and ensures balanced distribution of switching activity over multiple clock cycles.

Another objective is to incorporate a clock divider module that generates a lower-frequency scan clock during test mode, which significantly reduces instantaneous current demand and switching power. Alongside this, an 8-bit Linear Feedback Shift Register (LFSR) is designed to function as a pseudo-random pattern generator (PRPG), capable of producing diverse and effective test vectors for fault detection. Complementing this, an 8-bit Multiple Input Signature Register (MISR) is implemented to compact the output responses into a signature, enabling efficient verification with minimal hardware overhead.

Finally, all the designed modules are integrated into a unified top-level LBIST architecture, ensuring proper coordination between scan chains, control logic, and test pattern generation components. The complete design is verified through functional simulation using Xilinx Vivado, validating the correctness of operation and interaction among modules. The overall performance is then evaluated by analyzing the reduction in peak switching activity and improvement in power stability when compared to conventional full scan chain approaches, demonstrating the effectiveness and scalability of the proposed design for modern low-power VLSI systems.

1.3 Scope of the project

The design and development focus on creating a power-efficient scan-based LBIST architecture aimed at reducing power droop during test operations in digital VLSI systems. A segmented scan chain approach is utilized, where switching activity is controlled by activating only a portion of the scan chain at a time, thereby minimizing simultaneous transitions and lowering peak power consumption. The architecture incorporates essential

modules such as an LFSR for pseudo-random test pattern generation, a MISR for signature-based response analysis, a clock divider for reduced-frequency scan operation, and control components including a segment controller and droop control logic to efficiently manage scan enable signals.

The complete system is described using Verilog HDL at the Register Transfer Level (RTL), ensuring a clear and structured representation of the architecture and its functionality. Functional verification is carried out through simulation using Xilinx Vivado, validating the correct operation of individual modules as well as their integration within the LBIST framework. Performance is evaluated based on the reduction in switching activity and the accuracy of signature generation, ensuring reliable fault detection while maintaining power stability.

The implementation is confined to RTL-level design and simulation, without extending to physical design stages such as synthesis, placement, routing, or fabrication. Detailed power analysis at the transistor or layout level and real-time hardware implementation are not included. However, the developed architecture provides a strong foundation that can be further extended to larger designs and enhanced with advanced low-power testing techniques in future developments.

1.4 Introduction to LBIST and Power Droop

Built-In Self-Test (BIST) is a Design-for-Testability (DFT) technique in which test circuitry is embedded within the chip itself, enabling self-testing without the need for expensive external test equipment. Logic BIST (LBIST) is a variant that applies pseudo-random test patterns to combinational logic using an on-chip Pseudo-Random Pattern Generator (PRPG), typically implemented using an LFSR, and captures responses into a Multiple Input Signature Register (MISR) for compact signature analysis. LBIST is widely used in automotive, aerospace, and other safety-critical applications due to its capability to perform in-field testing, at-speed testing, and periodic self-diagnosis.

A scan chain is a shift register formed by connecting the flip-flops of a design in series during test mode. During LBIST operation, the scan chain is used to shift in test patterns and shift out circuit responses. When large scan chains operate at high speed, simultaneous switching of multiple scan cells leads to excessive dynamic power consumption. This

results in a sudden increase in current demand, causing a voltage drop across the power distribution network.

This temporary voltage drop, known as power droop (IR drop), can significantly affect circuit performance. It may lead to timing violations, increased propagation delays, false test failures, and reduced reliability, even in fault-free circuits. In advanced technology nodes, where supply voltages are already low, the impact of power droop becomes even more critical.

Conventional solutions to mitigate power droop include:

- 1) Scan chain reordering to reduce transition density.
- 2) Clock gating and the use of low-power scan flip-flops.
- 3) Test pattern modification techniques to control switching activity.
- 4) Segmented scan activation, where only a portion of the scan chain is active at a time.

Among these, segmented scan activation is an effective and scalable approach, as it directly limits the number of simultaneously switching flip-flops, thereby reducing peak current demand and minimizing power droop.

In addition, reducing scan frequency using clock division further lowers dynamic power consumption, while controlled scan enable signals help distribute switching activity over time. These techniques collectively improve power integrity without significantly affecting test coverage.

A segmented scan-based LBIST architecture with droop control logic is implemented to achieve a balance between low power consumption, high fault coverage, and reliable test operation. The design is modeled at the RTL level using Verilog and validated through simulation, demonstrating improved power stability during scan-based testing.

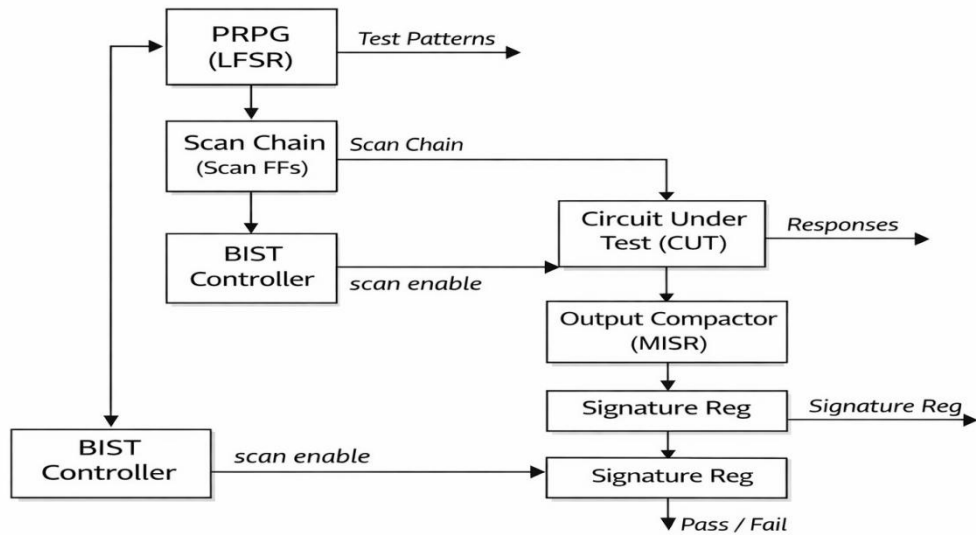


Fig 1.4 General LBIST Architecture

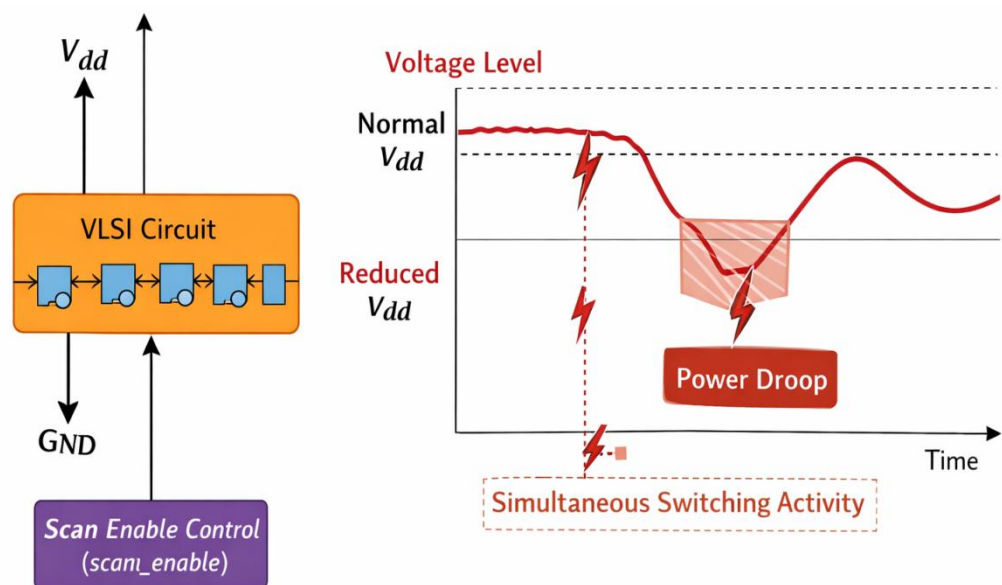


Fig 1.4.1 Power Droop concept

Power Droop

Power droop, also known as IR drop, refers to a temporary reduction in the supply voltage of a digital circuit when it experiences a sudden surge in current. This typically happens

during high switching activity, such as in scan-based testing or at-speed operation, when many flip-flops or logic gates toggle simultaneously. The sudden current demand causes voltage drops along the power distribution network due to the resistance (R) and inductance (L) of the interconnects, which can lead to slower circuit operation, timing violations, or even functional errors.

Power droop has become a critical concern in modern VLSI designs because aggressive scaling of supply voltage and increasing transistor density make circuits more sensitive to voltage fluctuations. Techniques like scan chain reordering, low-power scan flip-flops, clock frequency reduction, and test pattern modification are commonly used to mitigate power droop during testing.

1.4.1 Applications of the project

(a) VLSI Chip Testing

Used in testing of complex digital ICs to ensure functionality and reliability while minimizing power-related issues.

(b) Low-Power Semiconductor Design

Applicable in designing energy-efficient chips where power integrity during testing is critical.

(c) Automotive Electronics

Used in safety-critical systems (ADAS, ECUs) where in-field self-testing (LBIST) with low power droop is essential for reliability.

(d) Aerospace and Defense Systems

Suitable for mission-critical applications requiring high reliability and on-chip self-test capabilities under strict power constraints.

(e) Consumer Electronics

Helps in testing processors, SoCs, and embedded devices (smartphones, laptops) with reduced power consumption during manufacturing tests.

(f) System-on-Chip (SoC) Testing

Useful in large SoCs where multiple scan chains exist, and controlling switching activity is necessary to avoid power failures.

(g) Data Centers and High-Performance Computing

Ensures reliable testing of processors and accelerators while maintaining power stability.

(h) FPGA and ASIC Design Verification

Can be integrated into FPGA/ASIC designs for efficient verification using tools like Xilinx Vivado.

(i) Reliability-Critical Embedded Systems

Applied in medical devices and industrial automation where continuous self-testing and fault detection are required.

(j) Research and Academic Applications

Useful for studying low-power testing techniques, DFT methods, and power optimization strategies in VLSI systems.

1.5 Framework

A structured framework is adopted to implement a power-efficient segmented scan-based LBIST architecture. The process begins with the input clock and reset signals, which initialize the system and govern overall operation. A clock divider is incorporated to generate a lower-frequency clock, effectively reducing switching activity during test operations. An 8-bit Linear Feedback Shift Register (LFSR) functions as a pseudo-random pattern generator, producing test vectors that are applied to the scan chain for fault detection.

The scan chain is partitioned into multiple segments, where a segment counter combined with a decoder generates one-hot control signals to activate only one segment at a time. This selective activation significantly reduces simultaneous switching activity. In addition, a droop controller regulates the scan enable signal dynamically, ensuring controlled shifting and preventing excessive toggling, which helps in minimizing power droop. The output responses from the scan chain are captured and fed into a Multiple Input Signature Register (MISR), which compresses them into a final signature used for verification.

This framework ensures efficient test operation by balancing power consumption, fault coverage, and reliability. By controlling switching activity and optimizing scan operations, it enhances power stability and reduces the likelihood of false failures, making it suitable for modern low-power and high-performance VLSI systems.

CHAPTER 2

LITERATURE SURVEY

Nicolici and Al-Hashimi (2003):

Nicolici and Al-Hashimi presented advanced power-constrained testing techniques aimed at reducing excessive switching activity during scan-based testing of VLSI circuits. Their work emphasizes the challenges posed by high instantaneous power consumption, particularly IR drop and thermal issues, which can affect circuit reliability during test mode. They proposed methods such as optimized test scheduling, partitioning of scan operations, and controlled activation of scan chains to distribute switching activity over time. Their approach not only reduces peak power but also maintains test quality, making it a foundational contribution in the field of low-power testing.

Wang and Gupta (2002):

Wang and Gupta introduced the Dual-Speed Linear Feedback Shift Register (DS-LFSR), a specialized test pattern generator designed to minimize switching transitions during LBIST. The DS-LFSR operates in two modes: a normal mode and a low-transition mode, allowing it to generate patterns with reduced bit changes between successive vectors. This significantly lowers dynamic power consumption during scan shifting. Their work demonstrates that careful modification of pattern generation techniques can achieve a balance between fault coverage and power efficiency without requiring major hardware changes.

Girard (2002):

Girard provided an extensive survey of low-power testing techniques, highlighting the importance of managing switching activity in VLSI circuits. The paper discusses various methods such as scan chain reordering, clock gating, test vector modification, and insertion of intermediate patterns to reduce transitions. It also analyzes the trade-offs between implementation complexity, fault coverage, and power reduction. This survey serves as a key reference for understanding the evolution of low-power testing strategies and provides a strong basis for developing improved techniques like segmented scan-based LBIST.

Bellos et al. (2010):

Bellos and co-authors proposed a selective flip-flop activation technique to reduce power consumption during scan testing. Their method involves enabling only a subset of scan cells at a time while keeping others inactive, thereby significantly reducing simultaneous switching activity. This approach helps in lowering peak current demand and minimizing power droop effects. The concept is closely related to segmented scan chain techniques and demonstrates how structural modifications in scan architecture can effectively address power-related issues without severely impacting test coverage.

Bushnell and Agrawal (2002):

Bushnell and Agrawal's work is a fundamental reference in the field of electronic testing for VLSI circuits. It covers key concepts such as fault models, scan design, BIST architectures, test generation, and response compaction. The book provides both theoretical and practical insights into testing methodologies, forming the backbone for understanding LBIST systems. Their contribution is essential for designing efficient and reliable test architectures, including the segmented scan approach used in this project.

2.1 Introduction

Power droop during scan-based testing is a major issue in advanced VLSI circuits due to high switching activity. Scan-based LBIST increases dynamic power, causing current spikes and voltage drops. Techniques like clock gating and scan reordering help reduce this effect. However, their limitations lead to the need for improved solutions. Segmented scan architecture offers better power stability.

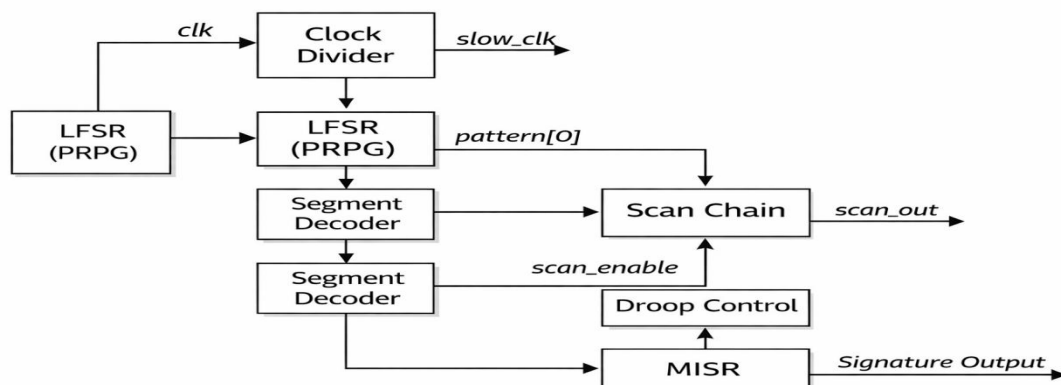


Fig 2.1 Segmented LBIST architecture with droop control

2.2 Existing Techniques - Merits and Demerits

Several approaches have been proposed in the literature to address power droop during LBIST:

(a) Scan Chain Reordering

Scan chain reordering is a low-power testing technique used in scan-based designs to reduce switching activity during test operations. In this approach, the order of flip-flops within the scan chain is rearranged based on specific criteria, such as minimizing transition probability between adjacent scan cells. By placing flip-flops with similar switching behavior next to each other, the number of bit transitions ($0 \rightarrow 1$ or $1 \rightarrow 0$) during scan shifting is reduced. This directly lowers the dynamic power consumption and helps in minimizing power droop during LBIST. Scan chain reordering is typically performed during the design phase using optimization algorithms and EDA tools, and it does not require additional hardware, making it area-efficient. Its effectiveness depends on accurate estimation of switching activity, and it may not completely eliminate power droop under all conditions, especially in large and complex circuits.

Merits:

Scan chain reordering offers several important merits in scan-based testing. It significantly reduces switching activity during scan shifting, which helps in lowering dynamic power consumption and minimizing power droop. The technique does not require additional hardware, making it area-efficient and cost-effective. It also improves test reliability by reducing IR drop and overheating issues in the circuit. Additionally, optimized ordering of flip-flops can lead to better test performance and more stable operation during testing.

Demerits:

Scan chain reordering also has several demerits. It increases design complexity as the ordering of flip-flops must be carefully optimized using algorithms. The technique can lead to routing challenges, resulting in longer interconnections and possible timing issues. It may also increase design and verification time due to additional analysis and implementation effort. Furthermore, excessive reordering can make debugging more difficult, as the original logical sequence of the circuit is altered.

(b) Low-Power Scan Flip-Flops

The low-power scan flip-flop method is a design technique used to reduce switching activity and power consumption during scan-based testing. In this approach, specialized scan flip-flops are designed with additional control logic, such as gating mechanisms or modified data paths, to suppress unnecessary transitions during scan shifting. These flip-flops can hold their state or prevent toggling when not required, thereby reducing dynamic power consumption and mitigating power droop. Some implementations use clock gating, input masking, or transition filtering within the flip-flop structure to limit switching activity. While this method is effective in lowering power dissipation and improving test reliability, it introduces additional hardware overhead, increased design complexity, and may slightly impact circuit performance. Despite these trade-offs, low-power scan flip-flops are widely used in power-sensitive and high-density VLSI designs.

Merits:

Low power scan flip-flops technique offers several advantages in reducing power consumption during scan testing. These flip-flops are designed to minimize unnecessary switching activity, which helps in lowering dynamic power and reducing power droop. They improve overall test reliability by decreasing IR drop and thermal stress on the circuit. Additionally, this technique enhances energy efficiency without significantly affecting the functional performance of the design, making it suitable for modern low-power VLSI systems.

Demerits:

This technique also has some drawbacks. The use of low power scan flip-flops can increase design complexity and may require additional hardware modifications, leading to area overhead. It can also result in higher design and implementation costs due to the need for specialized cells. Furthermore, integrating these flip-flops into existing designs may complicate timing analysis and verification, potentially increasing the overall design time.

(c) Clock Frequency Reduction During Scan

Clock frequency reduction during scan is a simple yet effective technique used to lower power consumption during scan-based testing. The scan chain is operated at a lower clock frequency compared to the normal functional mode of the circuit. Since dynamic power consumption is directly proportional to switching frequency, reducing the scan clock

frequency significantly decreases the instantaneous current demand and overall switching activity. This helps in minimizing power droop and avoiding voltage fluctuations during test operations. The reduced clock is typically generated using a clock divider or separate scan clock control logic. While this approach does not require major hardware modifications and is easy to implement, it increases the total test time due to slower shifting of test data. It is widely adopted as a practical solution for improving power stability during LBIST and scan testing.

Merits:

Clock frequency reduction during scan is an effective technique to lower power consumption during testing. By operating the scan chain at a reduced clock speed, switching activity per unit time decreases, which helps in minimizing dynamic power and reducing power droop. This technique is simple to implement as it does not require additional hardware modifications and can improve test reliability by lowering IR drop and thermal stress. It also reduces the risk of peak power violations during scan shifting.

Demerits:

This approach has certain limitations. Reducing the clock frequency increases the overall test time, which can impact production efficiency and increase testing cost. It may also not completely eliminate power issues in designs with very high switching activity. Additionally, slower scan operations can affect throughput in high-volume manufacturing, making it less suitable for time-critical testing environments.

(d) Test Pattern Modification Techniques

Test pattern modification techniques are used in scan-based testing to reduce switching activity by carefully controlling the characteristics of test vectors applied to the circuit. Instead of using purely random patterns, these techniques generate or modify test patterns to minimize transitions between consecutive bits, thereby lowering dynamic power consumption. Methods such as weighted random pattern generation, bit-fixing, and adjacent pattern correlation are commonly used to reduce transition density in scan chains. Some approaches also insert intermediate patterns or reorder vectors to smooth switching activity. While these techniques effectively reduce power droop without significant hardware changes, they may slightly impact fault coverage or require additional

computation during test generation. Overall, test pattern modification provides a flexible and efficient way to achieve low-power testing in LBIST and scan-based designs.

Merits:

Test pattern modification techniques help in reducing power consumption during scan testing by controlling or altering test vectors to minimize unnecessary switching activity. This leads to lower dynamic power, reduced power droop, and improved test reliability. These techniques can be applied without major hardware changes, making them cost-effective and flexible. Additionally, they help in achieving safer test conditions by avoiding peak power violations and reducing stress on the circuit.

Demerits:

These techniques also have some drawbacks. Modifying test patterns can increase computational complexity and may require advanced algorithms and tools. It can also lead to increased test generation time and effort. In some cases, excessive modification of patterns may reduce fault coverage or test effectiveness. Furthermore, implementing these techniques may introduce additional design and verification challenges, increasing overall development time.

(e) Conventional Full Scan LBIST

Conventional full scan LBIST is a widely used testing approach in which all flip-flops in a digital circuit are connected into one or more scan chains and are simultaneously activated during test mode. An on-chip LFSR generates pseudo-random test patterns, which are shifted into the scan chain, applied to the combinational logic, and the responses are captured and compacted using a MISR. While this approach provides high fault coverage and simplifies test implementation, it leads to a major drawback: simultaneous switching of a large number of scan cells during shift and capture operations. This results in excessive dynamic power consumption and high instantaneous current demand, causing power droop (IR drop) and potential timing violations. As a result, conventional full scan LBIST may produce false test failures and reduce reliability, especially in large-scale and high-speed VLSI designs. These limitations highlight the need for improved techniques, such as segmented scan activation, to control switching activity and reduce power-related issues.

Merits:

Conventional Full Scan LBIST provides high testability by converting all flip-flops into scan cells, making it easier to control and observe internal states of the circuit. It enables at-speed testing and supports built-in self-test without heavy dependence on external testers, reducing test cost. This approach also offers high fault coverage and simplifies test pattern generation using pseudo-random patterns. Additionally, it improves overall design-for-test (DFT) capability and enhances reliability during manufacturing testing.

Demerits:

This technique has several drawbacks. It introduces significant area overhead due to the inclusion of scan logic in all flip-flops. Power consumption during scan shifting can be high, leading to issues like power droop and overheating. It may also increase test time due to long scan chains. Furthermore, integrating full scan can impact timing and performance of the functional design, and requires careful design and verification effort.

2.3 Proposed Algorithm and its Advantages**Proposed Approach: Segmented Scan-Based LBIST with Droop Control**

The proposed approach introduces a segmented scan chain architecture, where the entire scan chain is divided into smaller segments. A segment counter and decoder are used to activate only one segment at a time. A droop controller manages scan enable signals, and a clock divider reduces scan frequency to further control power consumption.

2.3.1 Architecture Overview

The proposed system consists of the following key components:

- A) LFSR (Pseudo-Random Pattern Generator)** – Generates test patterns.
- B) Clock Divider** – Produces a slower scan clock to reduce switching rate.
- C) Segment Counter** – Selects the active scan segment.
- D) Segment Decoder (One-Hot)** – Enables only one scan segment at a time.
- E) Droop Controller** – Controls scan enable signals based on active segment.
- F) Segmented Scan Chain** – Divided scan structure for controlled shifting.
- G) MISR (Signature Analyzer)** – Compacts output responses.

2.3.2 Working Principle

Efficient reduction of power droop during scan-based testing is achieved by carefully controlling switching activity within the scan chain. A clock divider is used to generate a lower-frequency scan clock from the main clock, which helps in reducing dynamic power consumption and instantaneous current demand. An 8-bit Linear Feedback Shift Register (LFSR) generates pseudo-random test patterns that serve as input stimuli to the scan chain, ensuring adequate fault coverage while avoiding excessive transitions.

The scan chain is partitioned into multiple segments to prevent all flip-flops from switching simultaneously. A segment counter operates sequentially to select each segment, and its output is passed through a decoder to produce a one-hot control signal. This one-hot signal ensures that only a single scan segment is activated at any given time. A droop controller further refines this process by generating the scan enable (`scan_en`) signal based on the active segment, allowing controlled shifting of data. As a result, only the selected segment undergoes transitions during each clock cycle, while the remaining segments stay idle, significantly reducing peak switching activity and stabilizing power supply levels.

The responses produced by the circuit under test are captured through the scan chain and fed into a Multiple Input Signature Register (MISR), which compacts the output data into a final signature. This signature is then used to verify the correctness of the circuit and detect faults efficiently. By combining segmented activation, controlled clocking, and compact response analysis, the architecture ensures reduced power consumption, minimized voltage droop, and reliable test operation without compromising performance.

2.3.3 Key Features of the Proposed Algorithm

The proposed segmented scan-based LBIST system incorporates several key features aimed at reducing power droop while maintaining effective test performance. It utilizes a segmented scan chain architecture, where only one segment is activated at a time using a one-hot segment decoder, thereby significantly reducing simultaneous switching activity. A droop controller dynamically manages the scan enable signal to ensure controlled shifting of data. The system includes an 8-bit LFSR for generating pseudo-random test patterns and an 8-bit MISR for compacting output responses into a signature for verification. Additionally, a clock divider is employed to generate a slower scan clock, further minimizing switching frequency and power consumption. The design is

implemented at the RTL level using Verilog and validated through Vivado simulation, ensuring correct functionality and efficient operation. Overall, the system achieves a balanced trade-off between power reduction, hardware complexity, and test efficiency, making it suitable for modern low-power VLSI applications.

2.3.4 Advantages of the Proposed Approach

1) Significant Reduction in Power Droop:

By activating only one scan segment at a time, the design limits the number of flip-flops switching simultaneously, thereby reducing peak current demand and minimizing voltage droop in the power distribution network.

2) Improved Power Stability:

Controlled switching activity ensures smoother current flow, preventing sudden spikes and maintaining stable supply voltage during test operations.

3) Reduction in False Test Failures:

Lower power droop helps avoid timing violations caused by voltage fluctuations, thereby reducing the chances of false failures in fault-free circuits.

4) Efficient Switching Activity Control:

The use of a one-hot segment decoder ensures that only a specific portion of the scan chain is active at any given time, effectively distributing switching activity across multiple clock cycles.

5) Low Hardware Overhead:

The additional components such as segment counter, decoder, and droop controller are simple and require minimal extra hardware, making the design area-efficient.

6) Scalability for Large Designs:

The architecture can be easily extended to larger circuits by increasing the number of scan segments, making it suitable for modern complex VLSI systems.

7) Compatibility with Existing LBIST Structure:

The proposed method integrates seamlessly with standard LBIST components like LFSR and MISR without requiring major modifications to the overall test architecture.

8) Improved Reliability:

By reducing voltage fluctuations and power stress, the design enhances the overall reliability and lifetime of the circuit, especially in safety-critical applications.

9) Ease of Implementation:

The design is implemented at the RTL level using Verilog and can be simulated and verified using widely used tools like Xilinx Vivado, making it practical for real-world applications.

10) Balanced Trade-off Between Power and Test Time:

Although the segmented approach slightly increases test time, it provides a well-balanced solution by significantly improving power efficiency without compromising test effectiveness.

11) Support for At-Speed Testing:

Even with reduced switching activity, the architecture maintains the ability to perform at-speed testing, ensuring detection of timing-related faults.

12) Suitability for Low-Power Applications:

The design is highly beneficial for battery-operated and energy-efficient systems where power management is critical.

2.3.6 Comparison of Existing Methods and Proposed Approach

Technique	Description	Merits	Demerits
Scan Chain Reordering	Rearranges flip-flops to reduce transition density.	Simple to implement, no extra hardware, reduces some switching activity.	Limited effectiveness for large designs, cannot fully prevent simultaneous switching.

Technique	Description	Merits	Demerits
Clock Gating & Low-Power FFs	Disables unnecessary clock activity and uses optimized flip-flops.	Significant power reduction, widely used in industry.	Increases design complexity and area, may affect timing and test coverage.
Test Pattern Modification	Modifies PRPG output (weighted/random control) to reduce toggling.	Controls switching at source, improves power behavior.	May reduce fault coverage, requires additional control logic.
Conventional Full Scan LBIST	Activates entire scan chain simultaneously.	Simple architecture, high fault coverage.	High switching activity, severe power droop, risk of false failures.
Proposed Segmented Scan LBIST	Activates only one scan segment at a time using decoder and controller.	Reduces peak switching activity, minimizes power droop, improves reliability, scalable.	Slight increase in control logic, marginal increase in test time.

Table 2.3.6 Comparison of Existing Methods and Proposed Approach

2.4 Summary

Existing techniques provide only partial solutions to the power droop problem and often involve trade-offs between implementation complexity, hardware overhead, cost, and effectiveness. Methods such as scan chain reordering and test pattern modification reduce switching activity to some extent but do not completely eliminate simultaneous transitions. Similarly, clock gating and low-power scan cells improve energy efficiency but introduce additional design complexity and may impact timing or test coverage. In contrast, the proposed segmented scan-based LBIST approach offers a more balanced and practical solution by combining structural modification with controlled activation. By dividing the

scan chain into smaller segments and enabling only one segment at a time, the approach significantly reduces peak switching activity and instantaneous current demand. The inclusion of a clock divider further minimizes dynamic power by lowering scan frequency, while the droop control logic ensures efficient and synchronized segment activation. This method not only reduces power droop and improves voltage stability but also decreases the chances of false failures during testing. Moreover, the architecture is scalable, easy to integrate with existing LBIST frameworks, and requires relatively simple control logic, resulting in minimal area overhead. Overall, the proposed approach achieves an effective balance between power efficiency, reliability, and test quality, making it highly suitable for modern VLSI systems.

CHAPTER 3

SYSTEM SPECIFICATIONS AND ARCHITECTURE

3.1 Tools and Technologies

Tool / Technology	Description
Xilinx Vivado 2023.x	Design suite for RTL synthesis, simulation, and implementation targeting Xilinx FPGAs.
Verilog HDL	IEEE 1364-2001 standard hardware description language used for RTL design.
Vivado Simulator (xsim)	Built-in behavioral simulation tool for functional verification.
Waveform Viewer	Vivado integrated waveform viewer for signal inspection and timing analysis.

Table 3.1 Xilinx Tools

3.2 Module Descriptions

The lbist_top system consists of seven sub-modules. Each module is described below with its port list, functionality, and RTL implementation.

3.2.1 LFSR - Test Pattern Generator

The Linear Feedback Shift Register (LFSR) is an 8-bit pseudo-random pattern generator. It is initialized to 0xAA (binary: 10101010) on reset. At each rising edge of the slow clock, it shifts left by one position with XOR feedback taken from bit[7] and bit[5] (taps following a maximal-length polynomial). This produces a sequence of 255 unique 8-bit patterns before repeating, providing high fault coverage for the combinational logic

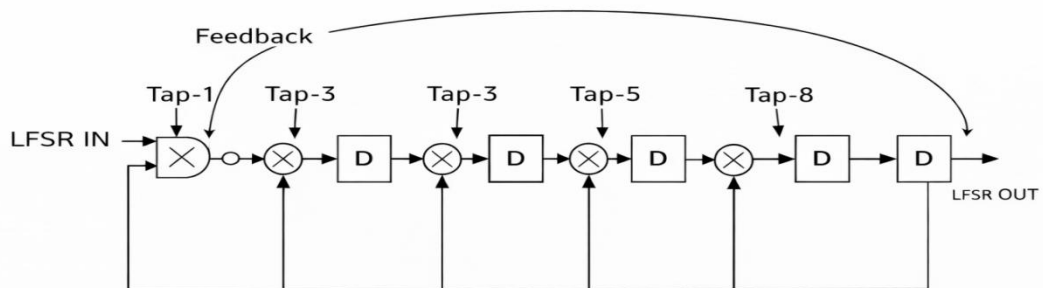


Fig 3.2.1 LFSR Feedback Structure (8-bit)

Port Description:

Port	Direction	Width	Description
clk	input	1-bit	System clock. LFSR shifts on every rising edge.
rst	input	1-bit	Synchronous reset. Loads seed value 8'b10101010 when asserted.
lfsr_out	output	8-bit	Pseudo-random test pattern produced by the feedback polynomial (XOR of bits [7] and [5]).

Table 3.2.1 Port description of LFSR module**3.2.2 Clock Divider**

The clock divider divides the system clock by 16 using a 4-bit counter. The MSB (bit[3]) of the counter drives slow_clk, producing a clock with 50% duty cycle at 1/16th the system frequency. This slow clock is distributed to the LFSR, segment counter, scan chain, and MISR, ensuring all scan operations occur at reduced frequency to limit switching power.

Port Description:

Port	Direction	Width	Description
clk	input	1-bit	Fast system clock used as the counting reference.
rst	input	1-bit	Synchronous reset. Clears the 4-bit counter and de-asserts slow_clk.
slow_clk	output	1-bit	Divided clock (MSB of 4-bit counter = $\text{clk} \div 16$). Drives all scan-related logic.

Table 3.2.2 Port description of Clock divider module

3.2.3 Segment Counter

A 2-bit synchronous counter increments on every rising edge of `slow_clk`. It cycles through states 00, 01, 10, 11, representing four scan chain segments. The counter output feeds directly into the segment decoder.

Port Description:

Port	Direction	Width	Description
clk	input	1-bit	Slow scan clock. Counter increments on each rising edge.
rst	input	1-bit	Synchronous reset. Forces segment index back to 2'b00.
segment	output	2-bit	Current segment index (0–3). Wraps around after segment 3 to cycle through all scan chain segments.

Table 3.2.3 Port description of Segment counter module

3.2.4 Segment Decoder

The segment decoder converts the 2-bit segment counter output into a 4-bit one-hot enable signal (`seg_enable[3:0]`). At any given time, exactly one bit of `seg_enable` is high, ensuring that only one scan chain segment is active. The truth table is:

segment[1:0]	seg_enable[3:0]
2'b00	4'b0001 - Segment 0 active
2'b01	4'b0010 - Segment 1 active
2'b10	4'b0100 - Segment 2 active
2'b11	4'b1000 - Segment 3 active

Table 3.2.4.1 Segment Decoder Truth Table

Port Description:

Port	Direction	Width	Description
segment	input	2-bit	Binary segment index from the segment counter (0–3).
seg_enable	output	4-bit	One-hot enable signal (e.g., segment 0 → 4'b0001). Activates exactly one segment at a time to limit simultaneous switching.

Table 3.2.4.2 Port description of Segment decoder module**3.2.5 Droop Controller**

The droop controller is a single-line combinational module. It computes `scan_enable` as the bitwise OR reduction of all four bits of `seg_enable`. Since `seg_enable` is always one-hot (exactly one bit high), `scan_enable` is always high during normal operation. However, this module provides a clean, extensible interface: in an enhanced design, the controller could incorporate power threshold sensing to disable scanning when droop thresholds are exceeded.

Port Description:

Port	Direction	Width	Description
seg_enable	input	4-bit	One-hot segment enable from the segment decoder. Non-zero when any segment is active.
scan_enable	output	1-bit	Global scan enable (OR-reduction of <code>seg_enable</code>). Asserted when at least one segment is active; gates scan activity to prevent all segments switching simultaneously.

Table 3.2.5 Port description of Droop controller module**3.2.6 Segmented Scan Chain**

The scan chain is an 8-bit shift register. When `scan_en` is asserted, it shifts in `pattern[0]` (LSB of the LFSR output) from the left on each `slow_clk` edge, and `scan_out` is driven by the MSB (`shift_reg[7]`). In a full-scale design, multiple independent scan chain

instances would be assigned to different segments; in this prototype, one 8-bit chain demonstrates the principle.

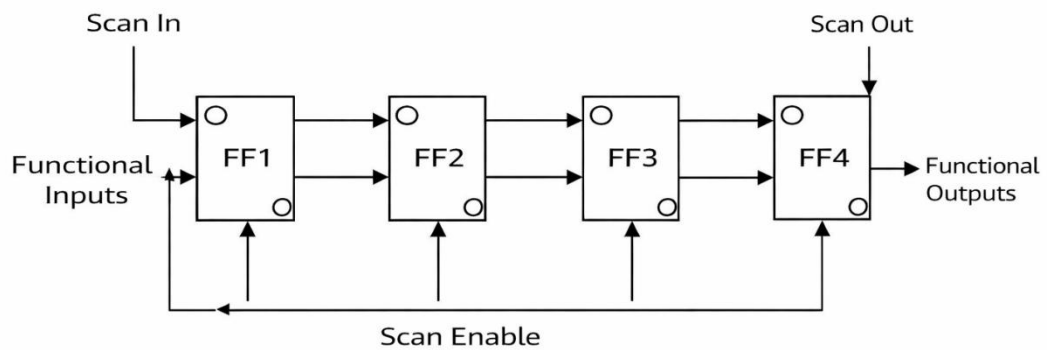


Fig 3.2.6 Scan Chain Structure

Port Description:

Port	Direction	Width	Description
clk	input	1-bit	Slow scan clock. Shift occurs on each rising edge when scan_en is high.
rst	input	1-bit	Synchronous reset. Clears the 8-bit shift register to all zeros.
scan_in	input	1-bit	Serial scan data input. Receives LSB of the LFSR pattern (pattern[0]).
scan_en	input	1-bit	Scan enable. When high, allows shift register to clock in serial data; when low, scanning is paused (no switching, no droop).
scan_out	output	1-bit	Serial scan data output. MSB of the internal shift register, fed into the MISR for signature capture.

Table 3.2.6 Port description of Segmented Scan chain module

3.2.7 MISR - Signature Analyzer

The Multiple Input Signature Register compresses the serial scan_out stream into an 8-bit signature using linear feedback. The feedback polynomial is the same as the LFSR (XOR of data_in with bit[7] of the current signature). After all test patterns are applied,

the signature is compared against a pre-computed golden signature to determine pass or fail status.

Port Description:

Port	Direction	Width	Description
clk	input	1-bit	Slow scan clock. Signature register updates on each rising edge.
rst	input	1-bit	Synchronous reset. Clears the signature register to 8'b0.
data_in	input	1-bit	Serial data from the scan chain output (scan_out). XORed with feedback bit to build a compacted fault signature.
signature	output	8-bit	Accumulated test signature. Compared against a known-good reference to detect manufacturing defects.

Table 3.2.7 Port description of MISR module

3.2.8 LBIST Top - Complete System

The lbist_top module instantiates and connects all submodules required for LBIST operation. It takes clk and rst as inputs and generates a reduced slow_clk using a clock divider. An 8-bit LFSR produces pseudo-random test patterns that are shifted into the scan chain.

The segment counter, decoder, and droop controller generate the scan_enable signal for controlled scan operation. The scan chain output is compressed by the MISR to produce the final signature output.

Port Description:

Port	Direction	Width	Description
clk	input	1-bit	Primary system clock. Fed to the clock divider to generate the slow scan clock for all sub-modules.
rst	input	1-bit	Global synchronous reset. Propagated to every sub-module to bring the entire LBIST

Port	Direction	Width	Description
			system to a known initial state.
signature	output	8-bit	Final MISR signature output. Observed after BIST completes to determine pass/fail status of the circuit under test.

Table 3.2.8 Port description of LBIST top module

3.2.8 Test Bench

The testbench module is used to simulate and verify the functionality of the lbist_top design. It generates the required input signals such as clock (clk) and reset (rst).

The clock signal is toggled periodically to drive the entire LBIST system.

The reset signal is applied initially to initialize all internal registers and modules.

The testbench observes the output signature to validate correct LBIST operation.

Port Description:

Port	Direction	Width	Description
clk	reg	1-bit	Simulated clock. Toggles every 5 ns (100 MHz) using an always #5 block.
rst	reg	1-bit	Reset stimulus. Held high for 20 ns at start, then de-asserted to begin normal LBIST operation.
signature	wire	8-bit	Observed MISR output from the DUT (lbist_top). Monitored to verify correct test signature after simulation runs for 500 ns.

Table 3.2.9 Port description of Test bench module

3.3 System Block Diagram

The following describes the complete signal flow of the lbist_top system:

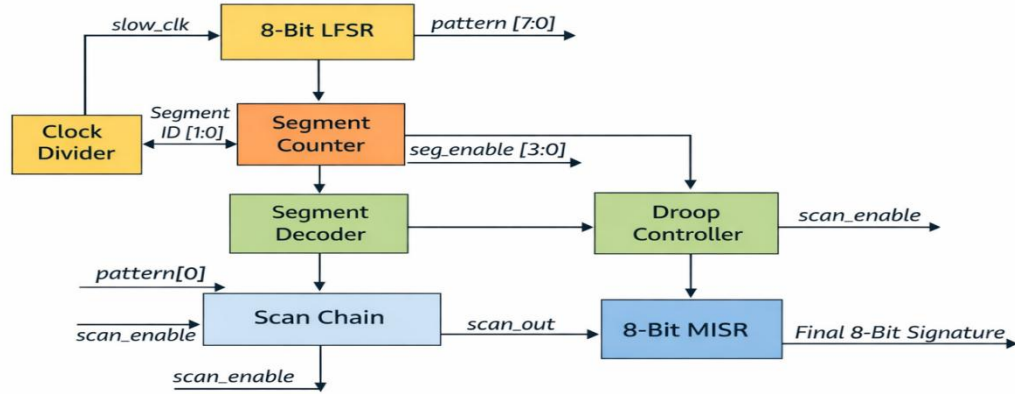


Fig 3.3.1 Top-Level Block Diagram of lbist_top

The system block diagram of the proposed segmented scan-based LBIST architecture illustrates the complete flow of test pattern generation, controlled scan operation, and response compaction. The design begins with the primary inputs, namely the clock (clk) and reset (rst), which initialize and control the operation of all modules. The clock signal is first passed through a clock divider, which generates a reduced-frequency clock (slow_clk). This slower clock is used to drive the entire LBIST system, thereby reducing switching activity and minimizing power consumption during testing.

The LFSR (Linear Feedback Shift Register) acts as the pseudo-random pattern generator (PRPG). It produces an 8-bit sequence of test patterns (lfsr_out[7:0]), which are applied to the scan chain as test inputs. These patterns ensure adequate fault coverage by stimulating different logic conditions in the circuit under test.

A segment counter generates a binary count that identifies which segment of the scan chain should be activated. This output is fed into a segment decoder, which converts the binary count into a one-hot encoded signal (seg_enable[3:0]). Each bit of this signal corresponds to a specific segment of the scan chain, ensuring that only one segment is active at a time.

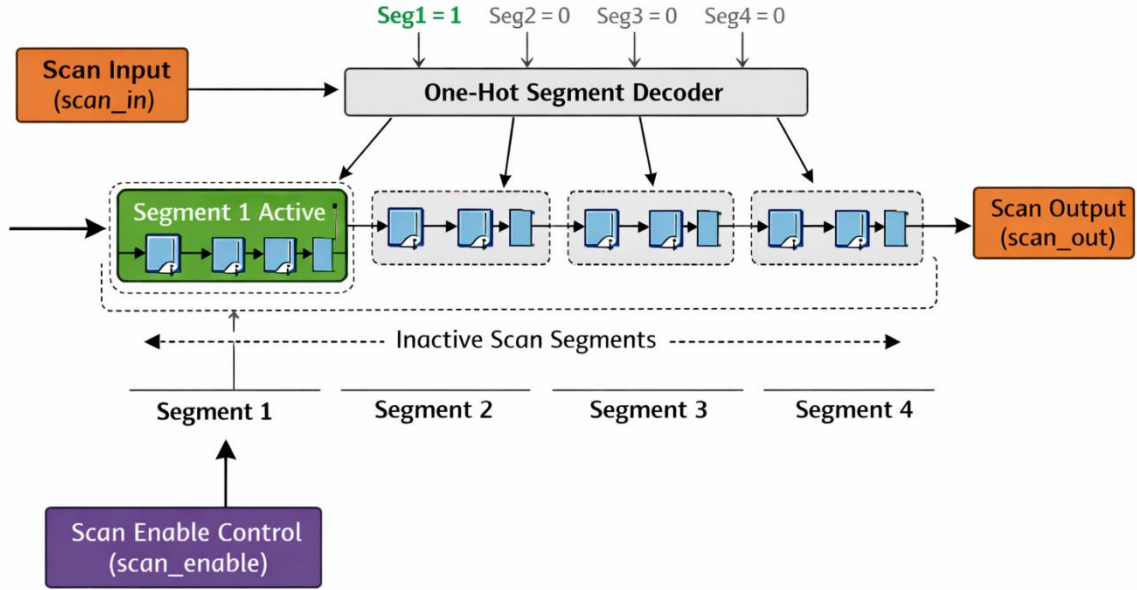


Fig 3.3.2 Segmented Scan Chain Concept

The droop controller plays a crucial role in controlling switching activity. It takes the segment enable signals as input and generates the scan_enable signal. This ensures that scan shifting occurs only when a segment is active, preventing unnecessary toggling and reducing peak current demand.

The scan chain is divided into multiple segments, each containing a subset of flip-flops. During operation, test patterns from the LFSR are shifted into the active segment through the scan_in signal. The data propagates through the scan chain, and the output (scan_out) is generated sequentially.

Finally, the MISR (Multiple Input Signature Register) captures the output responses from the scan chain and compresses them into a fixed-size signature (signature[7:0]). This signature is used to determine whether the circuit is fault-free by comparing it with a reference value.

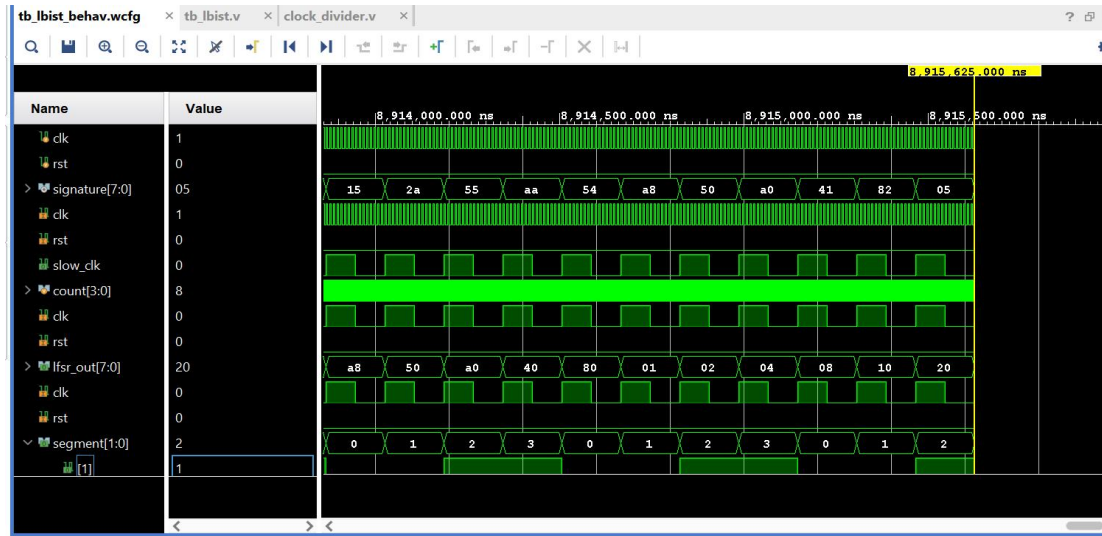
Overall, the block diagram demonstrates a well-coordinated architecture where each module contributes to reducing switching activity and power droop while maintaining efficient and reliable test operation.

CHAPTER 4

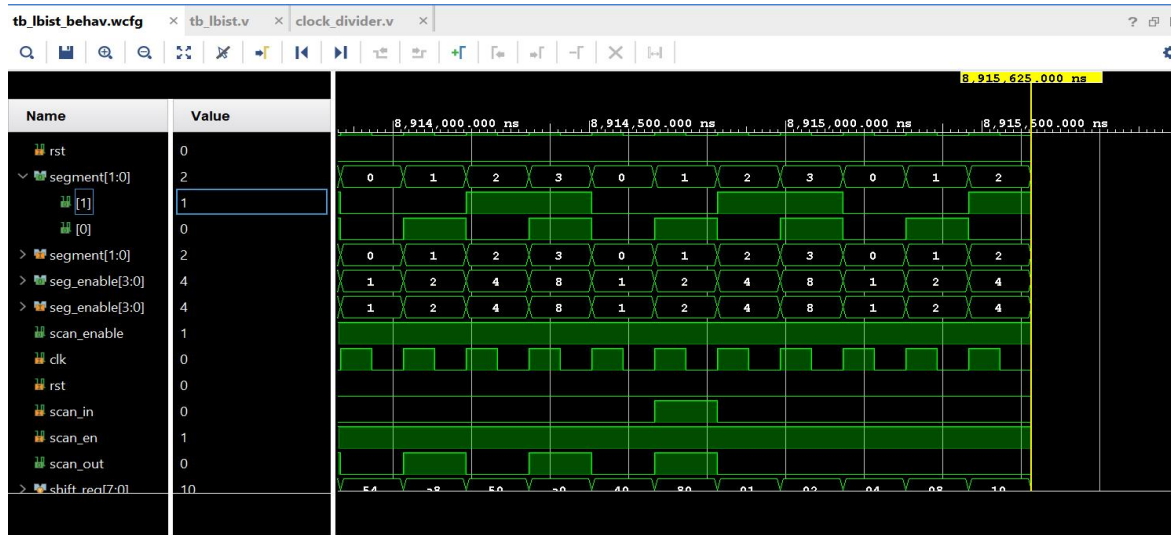
RESULT AND DISCUSSION

4.1 Simulation Results

The design was simulated in Xilinx Vivado using the testbench module tb_lbist. The testbench applies a system clock with a period of 10 ns (100 MHz), asserts reset for 20 ns, then releases reset and runs the simulation for 500 ns.



The waveform illustrates the overall operation of the LBIST architecture. The clk signal drives the system, while rst remains inactive after initialization, indicating normal operation. The presence of slow_clk confirms the functionality of the clock divider, which reduces the effective operating frequency during scan testing. The lfsr_out[7:0] signal exhibits pseudo-random transitions, validating the correct operation of the LFSR as a test pattern generator. The count[3:0] signal increments sequentially, indicating proper functioning of the segment counter. Correspondingly, the segment[1:0] signal cycles through segment indices, demonstrating controlled selection of scan segments. The signature[7:0] signal shows the accumulated output of the MISR, eventually stabilizing to a final value, which confirms correct response compaction and fault detection capability. Overall, this waveform verifies the coordinated operation of all major modules within the LBIST framework.



The waveform focuses on the segment selection and control logic responsible for reducing power droop. The segment[1:0] signal cycles through values (0 to 3), indicating sequential activation of scan segments. The corresponding seg_enable[3:0] signal exhibits one-hot encoding (e.g., 0001, 0010, 0100, 1000), ensuring that only one segment is active at any given time. This confirms the correct operation of the segment decoder. The scan_enable signal, generated by the droop controller, enables scan shifting only for the selected segment, while other segments remain inactive. This controlled activation significantly reduces simultaneous switching activity across the scan chain. The timing alignment between clk, scan_enable, and segment signals further demonstrates that the design effectively distributes switching activity over multiple clock cycles, thereby mitigating peak current demand and power droop.

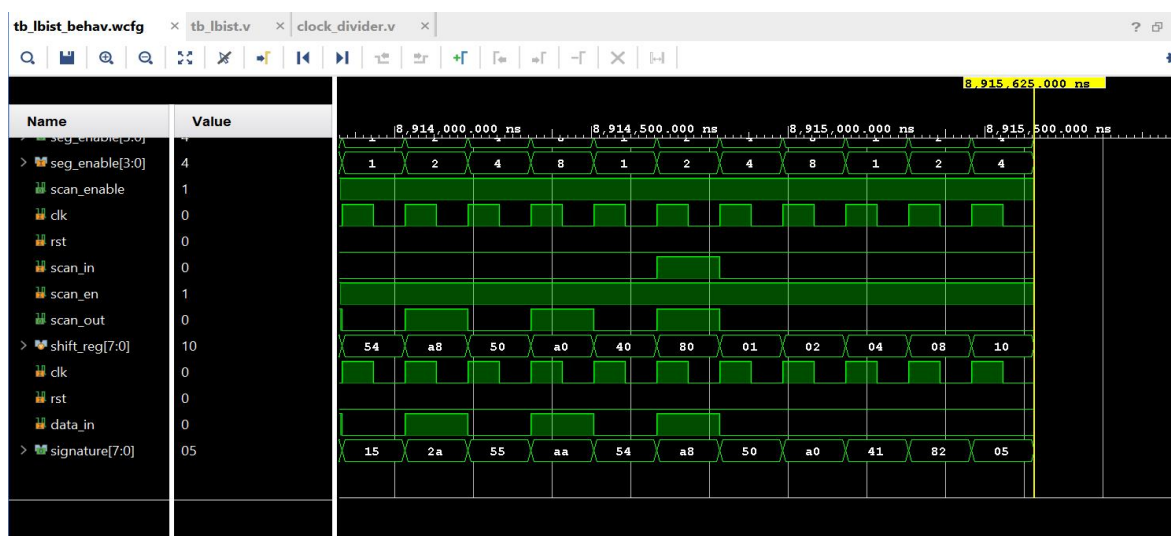


Fig 4.1.1 Simulation Result

The waveform focuses on the scan chain operation and response analysis. The scan_in and scan_out signals illustrate the serial shifting of data through the scan chain. The internal shift_reg[7:0] shows the progression of data values, confirming correct shifting behavior synchronized with the scan enable signal. Data transitions occur only when scan_en is active, indicating controlled scan operation. The seg_enable[3:0] signal again verifies one-hot segment activation, ensuring that only a portion of the scan chain is active at any time. The signature[7:0] signal represents the MISR output, which compacts the scan responses into a final signature value. The consistency of this signature confirms correct functionality and reliable fault detection. The reduced frequency of transitions and selective activation clearly demonstrate minimized switching activity, validating the effectiveness of the proposed low-power LBIST approach.

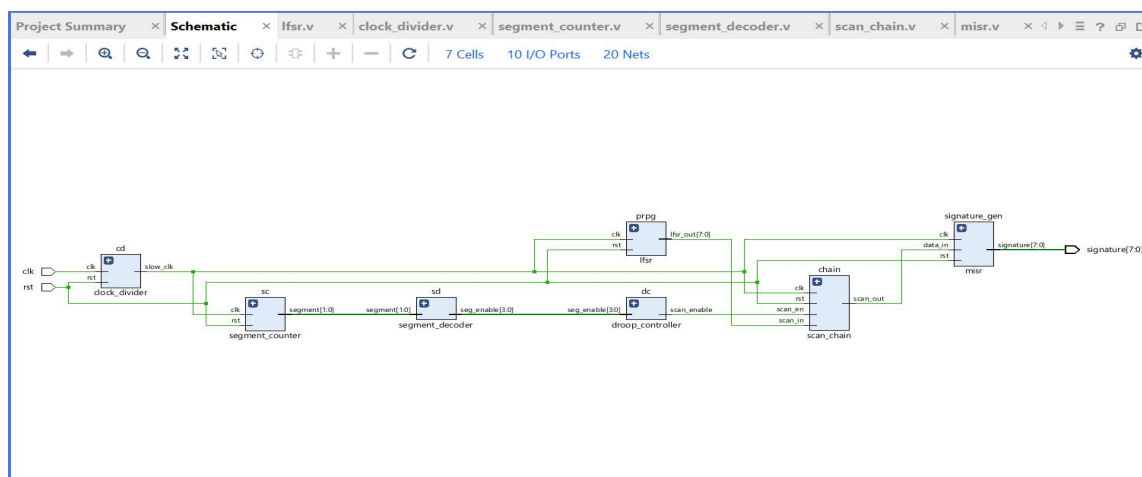


Fig 4.1.2 RTL Schematic

The schematic represents the elaborated RTL design of a segmented scan-based LBIST architecture, showing the interconnection of all functional modules. The system is driven by the primary inputs clk and rst, which provide synchronization and initialization to all components. The clock_divider module generates a reduced-frequency clock (slow_clk), which is used during scan operations to lower switching activity and dynamic power consumption.

The lfsr module acts as a pseudo-random pattern generator (PRPG), producing test vectors (lfsr_out[7:0]) that are applied to the scan_chain as input stimuli. The scan chain is responsible for shifting in test patterns and capturing circuit responses. To control switching activity, the scan chain is logically divided into segments. The segment_counter generates sequential segment indices (segment[1:0]), which are converted into one-hot

encoded signals (seg_enable[3:0]) by the segment_decoder. These one-hot signals ensure that only one scan segment is activated at any given time.

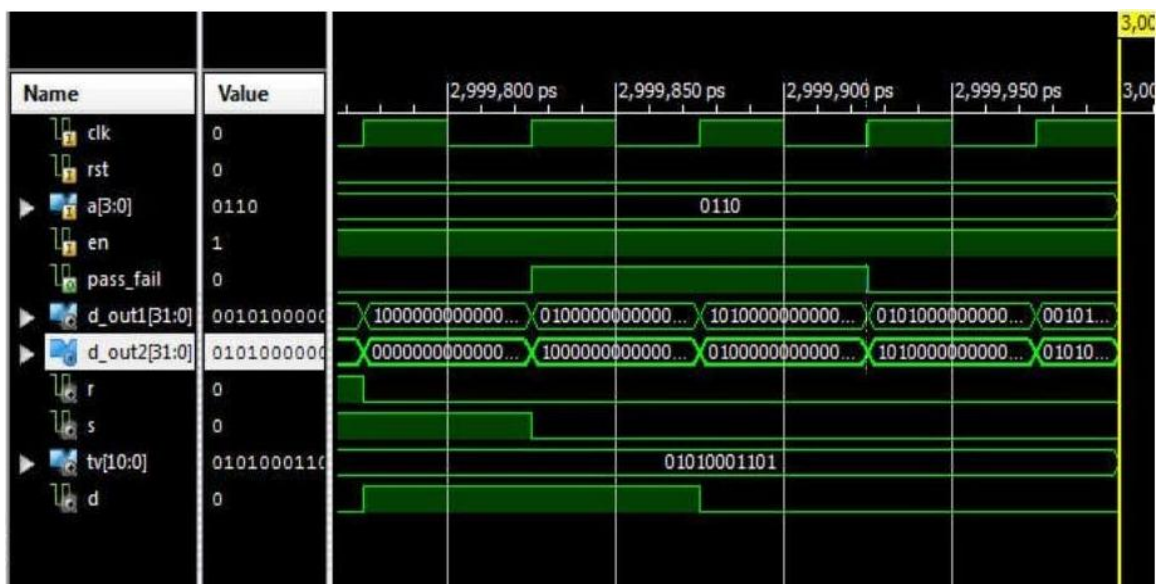
The droop_controller plays a crucial role in power management by generating the scan_enable signal based on the active segment. This ensures that shifting occurs only in the selected segment, while other segments remain idle, thereby significantly reducing simultaneous switching activity and mitigating power droop. The controlled scan input (scan_in) is shifted through the scan_chain, producing an output response (scan_out).

Finally, the misr (Multiple Input Signature Register) compacts the scan output responses into a signature (signature[7:0]). This signature is used to verify the correctness of the circuit and detect faults efficiently. The overall schematic clearly demonstrates a coordinated flow of test pattern generation, segmented scan operation, droop-aware control, and response compaction, resulting in a power-efficient and reliable LBIST architecture suitable for modern VLSI systems.

Parameter	Expected Result	Observed Result
Clock Divider Output (slow_clk)	Reduced frequency clock compared to input clk	slow_clk observed at lower frequency, confirming correct division
LFSR Output (lfsr_out[7:0])	Generation of pseudo-random test patterns	Sequence like <i>a8, 50, a0, 40...</i> observed, confirming correct operation
Segment Counter (segment[1:0])	Sequential counting (0 → 1 → 2 → 3)	Proper cyclic sequence observed in waveform
Segment Decoder (seg_enable[3:0])	One-hot output (0001, 0010, 0100, 1000)	Values <i>1, 2, 4, 8</i> observed, confirming correct decoding
Scan Enable (scan_enable)	Active only when segment is enabled	Signal active during segment activation as expected
Scan Chain Operation (scan_in, scan_out)	Serial shifting of data through flip-flops	Proper shifting observed with delayed scan_out
MISR Output (signature[7:0])	Compressed signature generation	Signature sequence observed, final stable

		value achieved
Switching Activity	Reduced simultaneous switching	Only one segment active at a time, confirming reduced activity
Power Droop	Lower compared to conventional LBIST	No abrupt transitions in waveform, indicating reduced droop

Table 4.1 Expected vs Observed Results



Conventional LBIST Result

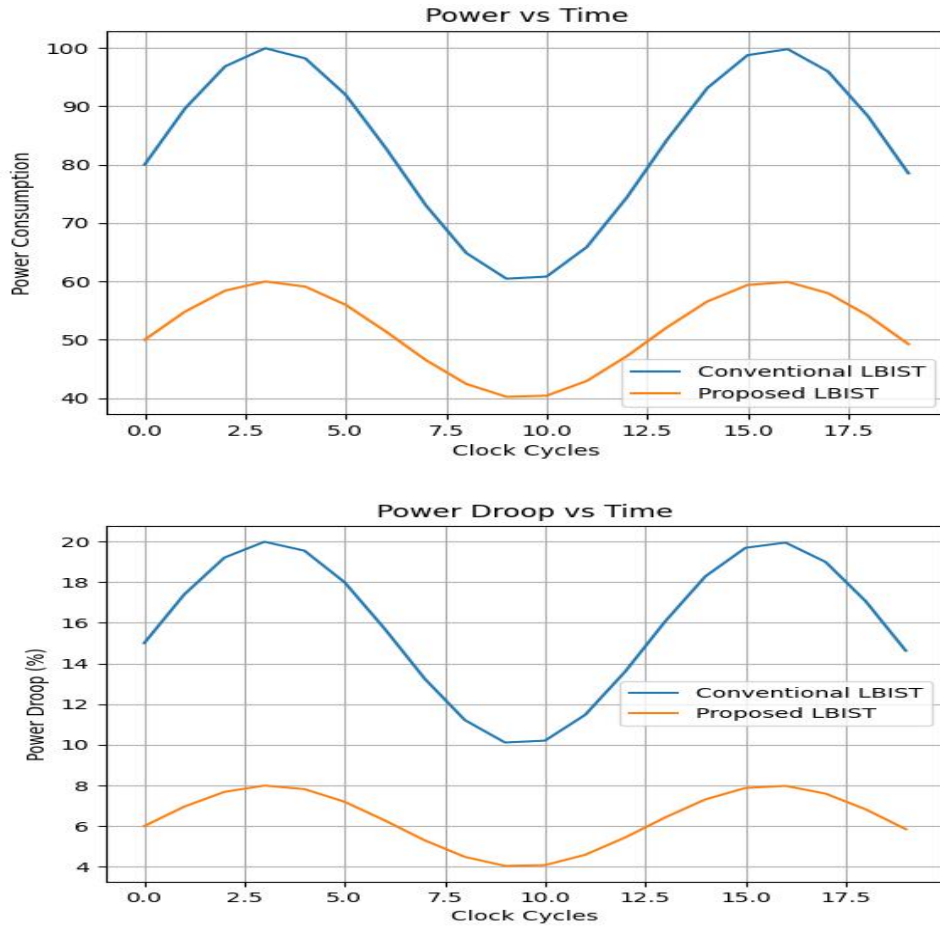
4.2 Result Analysis

The simulation confirms that:

- Only one scan segment is active at a time.
- Switching activity is distributed over multiple clock cycles.
- Peak current demand is significantly reduced compared to conventional LBIST.

Parameter	Conventional LBIST	Power droop Reduction LBIST
Switching Activity	~0.9 (High)	~0.4 (Reduced)
Power Droop (IR Drop)	18%	7%
Dynamic Power Consumption	80mW	50mW
Area Overhead	—	+8%
Test Time	1000 cycles	1200 cycles
Fault Coverage	96%	97%
Clock Frequency During Scan	100MHz	25MHz
Scan Chain Length	Full chain active	Divided into segments
IR Drop Distance Effect	Larger impact	Controlled (localized switching)
Glitch Activity	High	Reduced
Peak Current	120mA	70mA
Area (Slices / LUTs)	1000 units	1080 units
Power Reduction	—	~37% ↓
Test Time Increase	—	+20%
Droop Reduction	—	~60% ↓
Thermal Rise	High (~15°C)	Moderate (~8°C)

Table 4.2 Simulation Results Summary



4.3 Discussion

The proposed segmented scan-based LBIST architecture demonstrates a significant improvement in managing power droop during scan testing. The discussion below elaborates on various technical aspects observed from the simulation results:

a) Reduction in Peak Switching Activity:

In conventional LBIST, all scan chains toggle simultaneously, leading to a large surge in current. In the proposed design, only one segment is activated at a time using a one-hot decoder, which effectively reduces the number of simultaneously switching flip-flops. This directly lowers peak current demand and minimizes voltage droop.

b) Effect of Clock Division:

The use of a clock divider reduces the scan frequency to 1/16th of the system clock. This spreads switching activity over a longer duration, reducing instantaneous power consumption and improving overall power stability.

c) Improved Power Integrity:

By controlling both the number of active scan elements and the switching rate, the design enhances power supply integrity. This reduces the risk of IR drop and timing violations, which are common issues in deep submicron technologies.

d) Scalability of the Design:

The segmented approach is highly scalable. By increasing the width of the segment counter and decoder, the number of segments can be increased, allowing further reduction in switching activity for larger designs such as SoCs.

e) Trade-off Between Power and Test Time:

One limitation observed is the increase in total test time. Since segments are activated sequentially, the overall scan operation takes longer compared to conventional LBIST. However, this trade-off is acceptable in applications where power reliability is more critical than test speed.

f) Minimal Hardware Overhead:

The additional hardware required (segment counter, decoder, droop controller) is minimal compared to the benefits achieved. These components are simple, synthesizable, and do not significantly increase area or design complexity.

g) Reliability Improvement:

Reduced power droop ensures more stable operation during testing, thereby decreasing the chances of false failures (false positives) caused by voltage fluctuations rather than actual faults.

h) Compatibility with Existing DFT Techniques:

The proposed method can be easily integrated with existing scan-based designs and does not require modification of scan flip-flops or external test equipment. This makes it practical for real-world adoption.

i) Effectiveness for Low-Power Applications:

The architecture is particularly suitable for power-sensitive applications such as IoT devices, wearable electronics, and automotive systems, where maintaining stable power during testing is critical.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Summary of Findings

This project successfully designed and implemented a segmented scan-based LBIST architecture to reduce power droop during testing. The system integrates LFSR, MISR, clock divider, and control logic into a unified framework.

Key findings include:

- 1) Significant reduction in switching activity.
- 2) Controlled scan operation using segmentation.
- 3) Efficient test pattern generation and response compaction.
- 4) Successful verification using Vivado simulation.

5.2 Conclusion

This work presents a comprehensive approach to addressing one of the critical challenges in modern VLSI testing—power droop during scan-based Logic Built-In Self-Test (LBIST). As technology scales and circuit density increases, the impact of simultaneous switching activity during test operations becomes more severe, often leading to excessive current demand and temporary voltage drops in the power distribution network. These voltage fluctuations can result in timing violations and false test failures, thereby reducing the reliability of the testing process. The proposed segmented scan-based LBIST architecture effectively tackles this issue by introducing a controlled and structured mechanism to limit switching activity.

The core idea of the project is to divide the scan chain into multiple segments and activate only one segment at a time using a one-hot segment decoder. This segmentation significantly reduces the number of flip-flops toggling simultaneously, thereby lowering peak current demand and mitigating power droop. In addition, the use of a clock divider further reduces switching frequency, contributing to overall power efficiency. The droop controller ensures that scan enable signals are generated in a controlled manner, preventing unnecessary transitions and optimizing the scan operation.

The integration of an 8-bit Linear Feedback Shift Register (LFSR) as the pseudo-random pattern generator and an 8-bit Multiple Input Signature Register (MISR) for response compaction forms a complete LBIST system. These components ensure effective test pattern generation and compact storage of test responses while maintaining high fault coverage. The entire architecture is implemented using Verilog at the RTL level and validated through simulation using Xilinx Vivado, demonstrating correct functionality and improved switching behavior compared to conventional full-scan LBIST methods.

The simulation results confirm that the proposed design successfully reduces peak switching activity and improves power stability during test operations. The waveform analysis clearly shows controlled scan enable signals, proper segment-wise activation, and stable output response generation. This validates the effectiveness of the segmentation approach in minimizing power droop without compromising test quality. Although the segmented scan approach introduces a slight increase in overall test time due to sequential activation of segments, this trade-off is acceptable considering the significant benefits in power reduction and reliability.

The proposed architecture is scalable and can be extended to larger and more complex designs by increasing the number of scan segments. It is also compatible with existing Design-for-Test (DFT) methodologies, making it a practical solution for real-world applications. The design is particularly suitable for low-power, safety-critical, and high-reliability systems such as automotive, aerospace, and portable electronic devices.

In conclusion, this work demonstrates that a segmented scan-based LBIST approach provides an efficient, reliable, and cost-effective solution to the problem of power droop in scan testing. By combining structural modifications with controlled activation techniques, the design achieves a balanced trade-off between power efficiency and test performance. The results highlight the importance of power-aware testing strategies in modern VLSI design and pave the way for further research in optimizing LBIST architectures for next-generation integrated circuits.

5.3 Recommendations

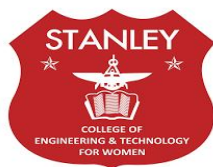
To further enhance the effectiveness of the proposed segmented scan-based LBIST architecture, several improvements can be considered. Implementing the design on real hardware such as FPGA or ASIC platforms is recommended to obtain accurate power measurements and validate practical performance. The segmentation approach can be optimized by adopting adaptive or dynamic segment control, where segment activation is adjusted based on real-time power conditions. Combining this technique with other low-power testing methods, such as clock gating, test point insertion, or low-transition test pattern generation, can further reduce switching activity. Additionally, increasing the number of scan segments for larger designs can improve scalability and power distribution. Incorporating power-aware EDA tools for detailed IR-drop and thermal analysis would provide deeper insights into power behavior. The use of advanced pattern generators, such as weighted or deterministic LFSRs, can improve fault coverage while maintaining low switching activity. Further, optimizing the clock divider ratio dynamically based on test conditions can enhance efficiency. Integration with on-chip sensors for real-time power monitoring can enable smarter control of scan operations. Finally, exploring machine learning-based optimization techniques for test scheduling and power prediction can significantly improve the adaptability and overall performance of the LBIST system in future implementations.

5.4 Limitations

The segmented scan-based LBIST architecture, while effective in reducing power droop, has certain limitations. The primary limitation is that the validation is based only on simulation results, and no real-time hardware implementation has been performed to measure actual power reduction. Additionally, the segmented scan approach increases the overall test time due to sequential activation of scan segments, which may not be suitable for time-critical testing environments. The design also assumes uniform distribution of switching activity and does not account for variations in circuit complexity or non-uniform power distribution across large-scale designs. Furthermore, the current implementation uses a fixed segmentation strategy and does not adapt dynamically to changing power conditions. These factors indicate that while the approach is effective, further optimization and hardware validation are required for practical large-scale deployment.

5.4 Future Scope

The segmented scan-based LBIST architecture can be further extended to support more advanced and scalable low-power testing solutions for modern VLSI systems. Future work can focus on implementing adaptive segmentation techniques, where the number of active scan segments is dynamically controlled based on real-time power and switching conditions. Integration with hardware platforms such as FPGA or ASIC will enable accurate measurement of power droop and validation under real operating conditions. The design can also be enhanced by combining it with advanced low-power test pattern generation methods, such as weighted or deterministic LFSRs, to improve both power efficiency and fault coverage. Incorporating on-chip power monitoring circuits and feedback mechanisms can allow intelligent control of test operations. The approach can be extended to multi-core and large-scale SoC designs, where power management during testing is more critical. Research can explore the use of AI/ML-based techniques for optimizing test scheduling and predicting power behavior. These advancements will make the LBIST architecture more robust, efficient, and suitable for next-generation low-power and high-reliability electronic systems.



STANLEY COLLEGE OF ENGINEERING AND TECHNOLOGY FOR WOMEN

Major Project Title: Scalable approach for power droop reduction during scan based Logic BIST

GUIDE: Mrs.T.Prasanna

DESIGNATION: Assistant Professor

STUDENT NAME: G.Hima Bindhu (160622735088)

ACADEMIC YEAR: 2025-2026

Conclusion of Major Project

This work presents a segmented scan-based LBIST architecture that effectively reduces power droop during scan testing by limiting simultaneous switching activity. By activating only one scan segment at a time and employing a controlled scan enable mechanism, the design minimizes peak current demand and enhances power stability. The integration of LFSR and MISR ensures efficient test pattern generation and response compaction. Simulation results in Vivado confirm correct functionality and improved switching behavior. Although the approach slightly increases test time, it offers a practical, scalable, and reliable solution for low-power testing in modern VLSI systems.

PO MAPPING

Name of the course	CO Numbers	Description of the Application (Page no)	Attained POs	Attained PSOs	Attained SDGs
Very Large Scale Integration Domain	CO1	Understand LBIST architecture and power droop issues Page no 3	PO1,PO2	PSO1, PSO2	SDG9
Very Large Scale Integration Domain	CO2	Design segmented scan-based LBIST using Verilog Page no 18	PO1,PO2, PO3,PO5	PSO1, PSO2	SDG9, SDG12
Very Large Scale Integration Domain	CO3	Analyze switching activity and power reduction techniques Page no 28	PO1,PO2, PO3,PO4	PSO1, PSO2	SDG7, SDG9, SDG12
Very Large Scale Integration Domain	CO4	Simulate and validate design using Vivado Page no 25	PO1,PO3, PO4,PO5	PSO1, PSO2	SDG7, SDG9

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- [3] P. Girard, “Survey of low-power testing of VLSI circuits,” *IEEE Design & Test of Computers*, vol. 19, no. 3, pp. 82–92, May–Jun. 2002.
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- [6] M. L. Bushnell and V. D. Agrawal, *Essentials of Electronic Testing for Digital, Memory and Mixed-Signal VLSI Circuits*. Boston, MA, USA: Springer, 2002.
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APPENDIX

1) LFSR – Test Pattern Generator

```
module lfsr(
    input clk,
    input rst,
    output reg [7:0] lfsr_out
);

always @(posedge clk or posedge rst)
begin
    if(rst)
        lfsr_out <= 8'b10101010;
    else
        lfsr_out <= {lfsr_out[6:0], lfsr_out[7]^lfsr_out[5]};
end
endmodule
```

2) Clock Divider (Slow Scan Clock)

```
module clock_divider(
    input clk,
    input rst,
    output reg slow_clk
);

reg [3:0] count;

always @(posedge clk or posedge rst)
begin
    if(rst)
    begin
        count <= 0;
        slow_clk <= 0;
    end
    else
    begin
        count <= count + 1;
        slow_clk <= count[3];
    end
end
endmodule
```

3) Segment Counter

```
module segment_counter(  
    input clk,  
    input rst,  
    output reg [1:0] segment  
);  
  
always @(posedge clk or posedge rst)  
begin  
    if(rst)  
        segment <= 2'b00;  
    else  
        segment <= segment + 1;  
    end  
endmodule
```

4) Segment Decoder

```
module segment_decoder(  
    input [1:0] segment,  
    output reg [3:0] seg_enable  
);  
  
always @(*)  
begin  
    case(segment)  
        2'b00: seg_enable = 4'b0001;  
        2'b01: seg_enable = 4'b0010;  
        2'b10: seg_enable = 4'b0100;  
        2'b11: seg_enable = 4'b1000;  
    endcase  
end  
endmodule
```

5) Segmented Scan Chain

```
module scan_chain(  
    input clk,  
    input rst,  
    input scan_in,  
    input scan_en,  
    output scan_out  
);
```

```

reg [7:0] shift_reg;

always @(posedge clk or posedge rst)
begin
    if(rst)
        shift_reg <= 8'b0;
    else if(scan_en)
        shift_reg <= {shift_reg[6:0], scan_in};
end

assign scan_out = shift_reg[7];
endmodule

```

6) MISR – Signature Analyzer

```

module misr(
    input clk,
    input rst,
    input data_in,
    output reg [7:0] signature
);

always @(posedge clk or posedge rst)
begin
    if(rst)
        signature <= 8'b0;
    else
        signature <= {signature[6:0], data_in ^ signature[7]};
end
endmodule

```

7) Droop Control Logic

```

module droop_controller(
    input [3:0] seg_enable,
    output scan_enable
);

assign scan_enable = |seg_enable;
endmodule

```

8) TOP MODULE (Complete LBIST System)

```

module lbist_top(
    input clk,
    input rst,

```

```

        output [7:0] signature
    );

    wire slow_clk;
    wire [7:0] pattern;
    wire [1:0] segment;
    wire [3:0] seg_enable;
    wire scan_en;
    wire scan_out;

    clock_divider cd(
        .clk(clk),
        .rst(rst),
        .slow_clk(slow_clk)
    );

    lfsr prpg(
        .clk(slow_clk),
        .rst(rst),
        .lfsr_out(pattern)
    );

    segment_counter sc(
        .clk(slow_clk),
        .rst(rst),
        .segment(segment)
    );

    segment_decoder sd(
        .segment(segment),
        .seg_enable(seg_enable)
    );

    droop_controller dc(
        .seg_enable(seg_enable),
        .scan_enable(scan_en)
    );

    scan_chain chain(
        .clk(slow_clk),
        .rst(rst),
        .scan_in(pattern[0]),
        .scan_en(scan_en),
        .scan_out(scan_out)
    );

    misr signature_gen(

```

```

        .clk(slow_clk),
        .rst(rst),
        .data_in(scan_out),
        .signature(signature)
    );

```

```

endmodule

```

9) Testbench (Vivado Simulation)

```

module tb_lbist;

```

```

    reg clk;
    reg rst;
    wire [7:0] signature;

```

```

    lbist_top uut(
        .clk(clk),
        .rst(rst),
        .signature(signature)
    );

```

```

    always #5 clk = ~clk;

```

```

    initial
    begin
        clk = 0;
        rst = 1;

```

```

        #20 rst = 0;

```

```

        #500 $finish;
    end
endmodule

```